

Inborn Errors of Amino Acid Metabolism

DR. WAJEEHA SHAMS ANSARI

Foundation of Inherited Metabolic Disorders



Single Gene Defects

Enzyme mutations causing total or partial loss of catalytic function.



Toxic Accumulation

Substrate accumulation causing intellectual disability and neurotoxicity.



Clinical Burden

Individual diseases rare (<1:250,000), but major pediatric burden overall.

Phenylketonuria (PKU): Pathophysiology

PAH Deficiency

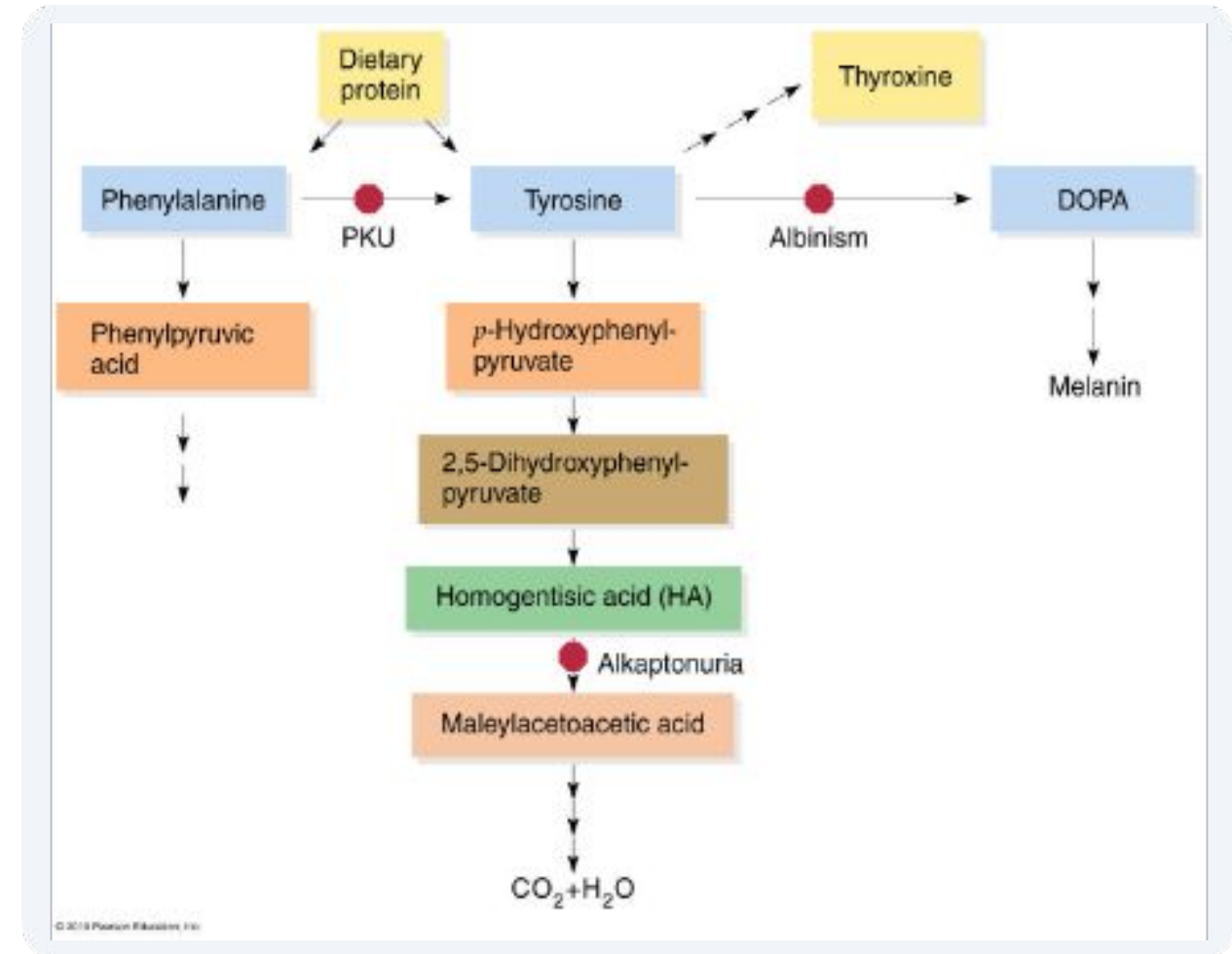
Primary defect in **phenylalanine hydroxylase** (1:15,000 incidence).

Conversion Block

Failure to convert Phenylalanine to Tyrosine causes hyperphenylalaninemia.

Essential Coenzyme

Requires **Tetrahydrobiopterin (BH₄)** and dihydropteridine reductase.



PKU: Manifestations & Screening

Elevated Metabolites

Phenylpyruvate and phenylacetate produce characteristic **mousy urine odor**.

CNS Deterioration

Untreated cases lead to severe intellectual disability (IQ < 50) and seizures.

Hypopigmentation

Inhibited tyrosinase reduces melanin, causing fair hair, blue eyes, pale skin.

Newborn Screening

Heel prick test at 24–48h post-protein feed detects blood Phe elevation.

PKU: Treatment Strategy & Maternal Risk

Lifelong Dietary Control

Synthetic Formula: Phe-free amino acids supplemented with tyrosine.

Diet Restriction: Strict avoidance of high-protein foods & aspartame.

Cognitive Protection: Early initiation within 7–10 days prevents IQ loss.

Maternal PKU Syndrome

Teratogenic Risk: High maternal blood Phe damages non-PKU fetus.

Congenital Malformations: Microcephaly and heart defects in newborn.

Pre-conception Diet: Dietary restriction mandatory prior to pregnancy.

Maple Syrup Urine Disease (MSUD)



BCKD Enzyme Defect

Deficient branched-chain α -ketoacid dehydrogenase blocks decarboxylation.



BCAA Accumulation

Leucine, Isoleucine, Valine accumulation causes maple syrup urine odor.



Therapeutic Intervention

Synthetic BCAA-limited formula; high-dose Thiamine (B₁) in variants.

Albinism: Tyrosine Metabolism Defect

Tyrosinase Deficiency

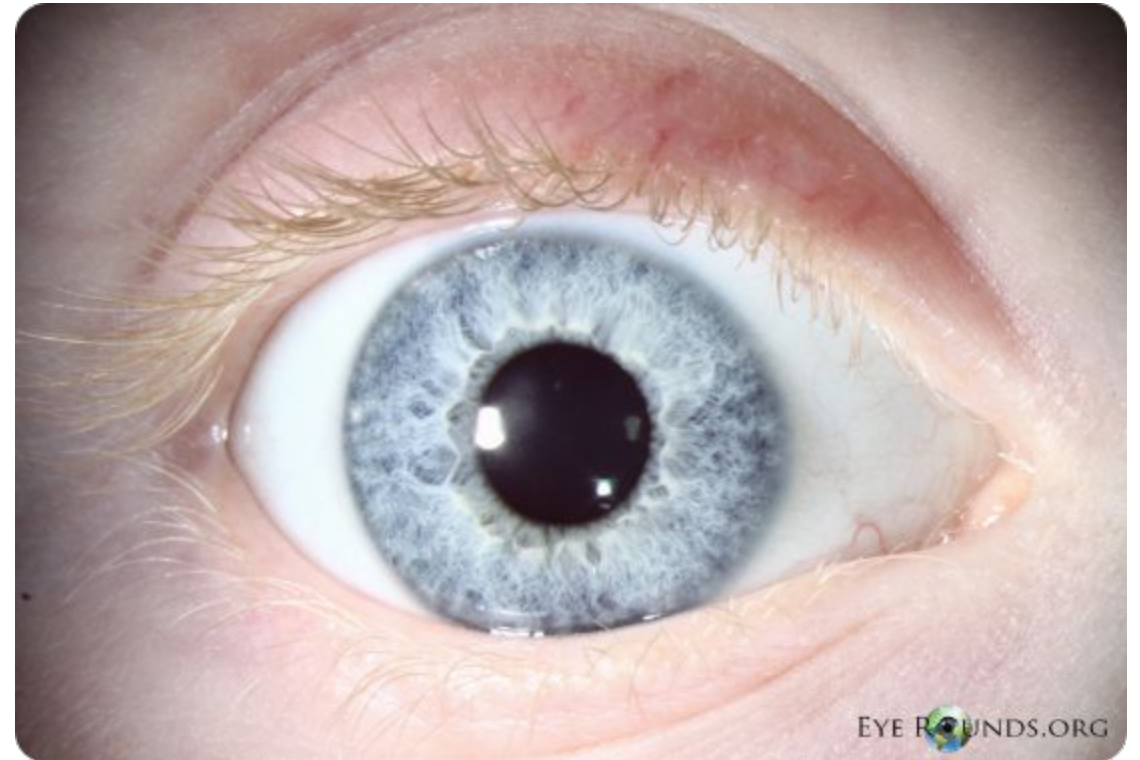
Absent or defective copper-requiring enzyme converts tyrosine to melanin.

Oculocutaneous Phenotype

Partial or complete hypopigmentation of skin, hair, and ocular iris.

Visual Pathologies

Associated photophobia, nystagmus, and high risk for skin carcinomas.



Homocystinuria: Transsulfuration Disorder

Enzyme Deficiency

Defect in **cystathionine β -synthase** converting homocysteine to cystathionine.

Clinical Features

Dislocated ocular lens (ectopia lentis), skeletal defects, and early thrombosis.

Targeted Therapy

High-dose Vitamin B₆ (pyridoxine) supplementation with methionine restriction.

Alkaptonuria & Related Conditions

Homogentisic Acid Oxidase Defect

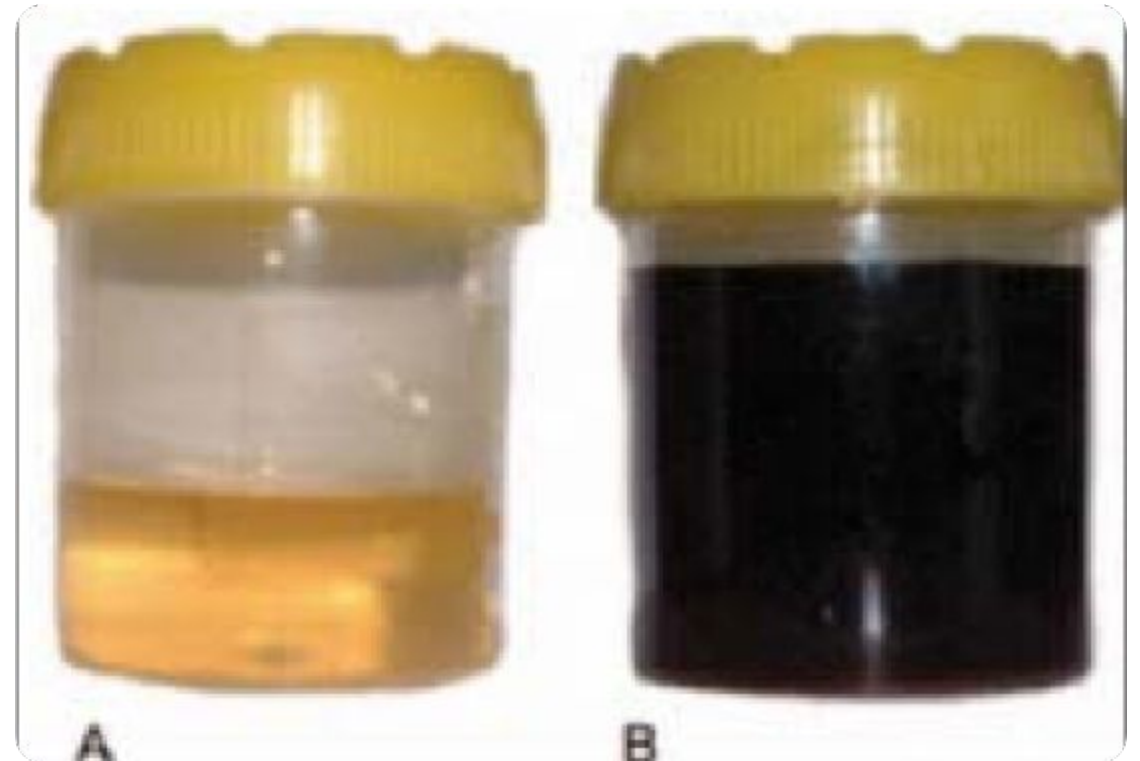
Causes accumulation of homogentisic acid in tyrosine degradation.

Homogentisic Aciduria

Urine turns black upon standing due to atmospheric oxidation.

Ochronosis & Arthritis

Dark pigment deposition in cartilage leading to severe joint destruction.



Clinical Matrix: Amino Acid IEM Comparison

Disorder	Deficient Enzyme	Key Accumulation	Diagnostic Sign
PKU	Phenylalanine Hydroxylase	Phenylalanine	Mousy odor, Low IQ
MSUD	BCKD Complex	Leucine, Isoleucine, Valine	Maple syrup urine odor
Albinism	Tyrosinase	Melanin deficit	Photophobia, white hair
Homocystinuria	Cystathionine β -synthase	Homocysteine	Lens dislocation, thrombosis
Alkaptonuria	Homogentisic Acid Oxidase	Homogentisic Acid	Urine turns black, ochronosis

